AUTHFi: Cross-Technology Device Authentication via Commodity WiFi

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***Abstract*—The explosive growth of the Internet of Things (IoT) has dramatically increased the demand for secure mechanisms to protect against unauthorized access and attacks. Traditionally, expensive Software-Defined Radios (SDRs) have been utilized to gather IoT physical features, which are critical for reliable authentication. However, the high cost of SDRs makes them impractical for widespread deployment across the vast and diverse IoT ecosystem. In contrast, this paper presents AUTHFi, a novel cross-technology device authentication framework that transforms the SDR approach for collecting and authenticating IoT device signals (e.g., ZigBee and Bluetooth) by utilizing commercial WiFi devices. Specifically, AUTHFi leverages the recent advances in Cross-Technology Communication (CTC) to reconstruct the partial waveform of IoT transmission, thus eliminating the requirement for expensive SDRs. AUTHFi requires us to address several unique challenges. First, AUTHFi compensates for signal losses of the partial waveform to get more signal information. Then, it introduces an enhanced Carrier Frequency Offset (CFO) estimation and a fusion neural network that combines CFO and the reconstructed waveform for accurate device authentication. We implement AUTHFi based on RTL8812au (commodity WiFi) and CC2652P (commodity ZigBee/Bluetooth). Our thorough evaluation confirms that AUTHFi offers reliable authentication under various settings, achieving a maximum accuracy of 94.2%.**

***Index Terms*—Device authentication, Cross-Technology Communication (CTC), physical-layer security**

I. Introduction

The proliferation of Internet of Things (IoT) significantly enriches our daily lives through many important applications, such as smart homes, healthcare monitoring and industrial automation [1]. However, these IoT devices are susceptible to various attacks, including exploitation of weak or default passwords, man-in-the-middle attacks, and replay attacks [2]. These attacks not only harm user privacy but also lead to potential economic losses [3].

Targeting this fundamental issue, encryption has been widely adopted. For example, AES-128, a symmetric-key encryption, is already a part of ZigBee and Bluetooth standards for secure communication [4]. However, these cryptography methods require significant computation, often too costly for low-end IoT devices such as ZigBee. According to [5], decryption requires over 88% of a ZigBee node's energy consumption.

To address the limitations of cryptographic solutions for resource-constrained IoT devices, recent techniques leverage the unique signal characteristics of different IoT nodes for device authentication. While these approaches offer high accuracy, they typically require dedicated Software-Defined Radios (SDRs) (e.g., Universal Software Radio Peripheral (USRP)) to collect In-phase and Quadrature (IQ) signals, followed by signal analysis. Note that while standard WiFi chips can estimate the CFO, this estimation is usually unavailable for external applications due to API limitations. For example, many techniques calculate the Carrier Frequency Offset (CFO), which is caused by the distinctive hardware imperfections, to uniquely identify the transmitter. This line of techniques has been implemented across various protocols such as ZigBee [6], Bluetooth [7], [8], LoRa [9], [10] and Wi-Fi [11], achieving notable success in mitigating security threats. Although this method reduces the computational load on IoT devices, it introduces additional complexity, energy consumption, and the costly inconvenience of deploying SDRs for wireless signal collection.

Fortunately, the emerging Cross-Technology Communication (CTC) solutions offer potential opportunities for authentication. CTC aims to enable direct communication between a wide range of wireless technologies without hardware modifications [12]–[14]. For example, XFi [15] enables commodity WiFi devices to reconstruct the partial waveform of ZigBee transmission and demodulate the transmitted ZigBee content.

Previously, IoT authentication methods typically necessitated the use of complex encryption mechanisms [16] or the deployment of dedicated SDRs to collect signals [17], imposing significant technical and financial burdens on the system infrastructure. Leveraging the recent advances in CTC, this paper introduces AUTHFi, which offers a more cost-effective and less complex solution for IoT authentication (e.g., Bluetooth and ZigBee) by utilizing standard WiFi devices

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Table I: Comparative Analysis of Authentication Methods

Metric	SDR-based Methods	Traditional Methods	AUTHFi
Device Cost	High ($1,000+ per USRP N210)	Low (No additional hardware)	Low (Uses existing WiFi)
IoT Energy Consumption	Low (Nomral signal transmission)	High (Extra encryption operation)	Low (Normal signal transmission)
Hardware Requirement	High (SDR installation at multiple points and extra device configuration)	Low (Software updates to ZigBee devices)	Low (Existing WiFi infrastructure and packet analysis tools)
Authentication Accuracy	High (Reliable in controlled environments)	High (Dependent on key management strength)	High (Good performance across scenarios)

instead of requiring specialized SDRs or complex encryption methods. However, developing AUTHFi is more than simply combining existing CTC and IQ-based authentication techniques. It requires resolving several key challenges as follows. First, AUTHFi uses existing CTC techniques, e.g., XFi [15], to reconstruct the partial waveform of IoT transmission, which suffer from signal losses due to WiFi's Low-Density Parity-Check (LDPC) coding and Cyclic Prefix (CP). Therefore, AUTHFi compensates for CP-related signal losses (0.8 μs) through interpolation, while longer LDPC losses cannot be reliably reconstructed due to their extended duration (3.2 μs). Second, AUTHFi enhances CFO estimation under the inevitable low quality of the reconstructed signal. Third, considering the potential fluctuations in the estimated CFO values due to factors such as signal attenuation and multi-path fading, AUTHFi introduces a robust authentication model that integrates IQ components and CFO values for accurate authentication. Fourth, AUTHFi enhances multi-device authentication efficiency through the allocation of frequency and temporal resources.

The major contributions of this paper are listed as follows:

1) We present AUTHFi, the first cross-technology device authentication framework. This framework allows commodity WiFi devices to authenticate IoT devices such as Bluetooth and ZigBee, facilitating ubiquitous authentication. In contrast, prior solutions commonly depend on the deployment of SDRs.
2) Aiming for cross-technology device authentication, we introduce a series of novel designs, including signal compensation, enhanced CFO estimation, a fusion neural network, and efficient multi-device coordination. These innovations enable a flexible and secure authentication framework by leveraging the strengths of CTC.
3) We implement AUTHFi on standard WiFi (RTL8812au) and ZigBee/Bluetooth devices (CC2652P). The evaluation results demonstrate that AUTHFi provides reliable authentication across different environments, achieving a maximum accuracy of 94.2% with the IQ+CFO fusion network. This performance significantly surpasses other benchmarks (i.e., the individual IQ and CFO networks)

The remainder of this paper is organized as follows. Section II discusses the motivation for cross-technology device authentication. Section III details the background and challenges of our AUTHFi. Section IV and V gives an overview and threat model, while Section VI delves into AUTHFi's detailed design. Section VII and VIII presents the experimental evaluation and security analysis of AUTHFi. Section IX discusses the related work for physical-layer authentication and CTC. Finally, the paper concludes in Section X.

II. MOTIVATION

This section introduces the existing issues in IoT authentication, motivating us to design AUTHFi, a cross-technology device authentication that relies on commodity WiFi devices.

1) **Challenges in Resource-Constrained IoT:** For IoT device authentication, encryption has been widely adopted to safeguard data privacy. For example, both ZigBee and Bluetooth employ AES-128 encryption in their standards [18]. However, encryption poses significant cost requirements for low-end devices. For a ZigBee node, its decryption is time-intensive — the decryption of a 60-byte payload requires 35 ms, far more than the 3 ms reception time. As a result, the decryption alone consumes over 88% of energy consumption for a ZigBee node, making it challenging for battery-powered IoT devices [5]. Moreover, key distribution in IoT devices often relies on a trust center [19] for key generation and negotiation, leading to increased network interactions and energy consumption.
2) **Limitations of SDR-based Authentication:** As a remedy for encryption-based authentication, researchers propose to leverage SDRs for authentication. These techniques collect raw IQ samples at the physical layer and further analyze them for device identification. Although promising, they inevitably require a dedicated SDR [20], [21], which could cost thousands of US dollars [20], in addition to the deployment cost. These high costs directly limit the potential applicability of SDR-based approaches.
3) **Opportunities via CTC:** CTC is an emerging technique enabling direct information exchange between heterogeneous wireless technologies [14], [22]–[24], while requiring only software modification. For example, XFi [15] enables commodity WiFi devices to reconstruct the signals transmitted from various IoT technologies (e.g., ZigBee and LoRa) and further demodulate the transmitted content. These recent advances motivate us to repurpose existing CTC techniques for the authentication of IoT devices via Commercial-Off-The-Shelf (COTS) WiFi devices, thus mitigating the limitations of existing SDR-based approaches.

Table I provides a comparative analysis of traditional SDR-based methods, encryption-based methods, and the proposed AUTHFi framework, focusing on key metrics including device cost, IoT energy consumption, hardware requirement, and authentication accuracy.

Traditional SDR-based methods (e.g., [7]–[11], [25]–[28]) achieve high authentication accuracy through precise signal capture using expensive SDR hardware like USRP N210

($1,000+ per unit). While these methods minimize IoT device energy consumption by only requiring normal signal transmission, they demand significant deployment effort for SDR installation and configuration at authentication points.

Encryption-based methods (e.g., [4], [5]) offer a low-cost solution by utilizing existing IoT hardware. However, the additional encryption operations increase IoT device energy consumption. These methods require minimal deployment effort through software updates but may impact resource-constrained IoT device performance.

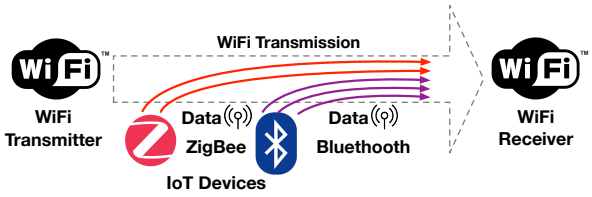
The proposed AUTHFi framework leverages existing WiFi infrastructure for authentication, eliminating additional hardware costs. It maintains low IoT energy consumption as devices operate normally without extra computation. With minimal deployment requirements and high authentication accuracy (up to 94.2%), AUTHFi provides an efficient and practical solution for IoT device authentication.

III. Background and Challenges

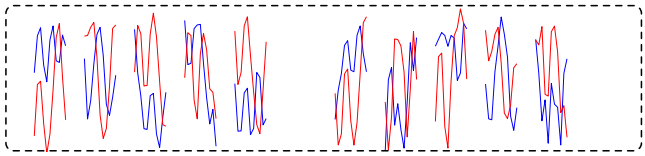
In this section, we first introduce the background of CTC, i.e., how to enable WiFi to receive signals from IoT devices. Then, we discuss the unique challenges in cross-technology device authentication.

A. Background

The recent advances in CTC enable direct communication between heterogeneous devices with only software modifications. For example, XFi [15] utilizes commodity WiFi cards to receive IoT transmissions (e.g., ZigBee and Bluetooth) as shown in Fig. 1(a). XFi employs a novel technique called signal hitchhiking, where IoT signals intentionally "hitchhike" on WiFi transmissions to be captured by WiFi receivers. First, a WiFi transmitter transmits a CTC packet, which could be demodulated by IoT devices and trigger IoT devices to transmit. During the ongoing WiFi transmission for CTC, IoT devices (e.g., ZigBee and Bluetooth) concurrently send their packets, thus affecting the received WiFi waveform and changing the received WiFi payload. Note that although other CTC approaches support direct IoT to WiFi communication, these studies [29]–[31] require modifications to the physical layer of WiFi. Therefore, we do not consider these methods here. Second, XFi analyzes the received WiFi payload through the following procedure: (1) XFi first obtains a sequence of new bits based on the received WiFi payload by reverse channel coding. (2) It then maps these bits to the frequency domain, specifically to WiFi subcarriers. (3) The final step involves performing an Inverse Fast Fourier Transform (IFFT) to reconstruct the IoT waveform in the time domain. Note that we assume ZigBee and Bluetooth are using coordinated channels during transmission to avoid interference. However, the reconstructed waveform suffers from signal losses due to WiFi's coding such as LDPC code and CP segments. Fig. 1(b) shows an example of the partial waveform reconstructed by XFi, while signal losses are marked by blank spaces. Finally, XFi analyzes this waveform to demodulate the transmitted message.



(a) Direct data transmission from IoT devices to commodity WiFi



(b) Partial waveform reconstruction

Figure 1: CTC from IoT to WiFi devices

B. Design Points

Based on the emerging opportunities offered by CTC, AUTHFi will move one step forward to enable cross-technology device authentication, which leverages commodity WiFi devices for authenticating IoT nodes such as Bluetooth and ZigBee. However, to achieve this goal, we must consider several key design points in existing CTC studies.

- **Partially Recovered Signals:** Existing CTC techniques, e.g., XFi, only reconstruct partial signals (e.g., Fig. 1(b)) of IoT transmissions due to the coding and CP in the WiFi standard. For CTC purposes, this issue could be addressed via additional coding. However, this signal loss imposes severe issues for physical layer-based authentication and demands the development of new compensation algorithms that discover hidden opportunities between heterogeneous wireless protocols [32].
- **Robust CFO Estimation:** In contrast to the signal captured by dedicated SDRs, the reconstructed IoT signal suffers from interference and noise, leading to distortions in the reconstructed waveform. To address this issue, it is essential to design a robust CFO estimation method that can effectively minimize these distortions, thereby enhancing the precision of CFO estimation.
- **Adaptive Authentication Model:** Although CFO is widely used for authentication, it has several limitations in practice. First, different IoT devices may have subtle CFO differences, making it difficult to distinguish certain devices. Second, the estimated CFO values can fluctuate due to environmental factors such as signal attenuation and multipath fading. However, it is important to note that the true CFO value is determined by the difference between two frequency oscillators and is only subject to physical factors such as temperature. As a result, accurate cross-technology device authentication calls for an adaptive authentication model with more reliable performance.
- **Concurrent Authentication:** Since WiFi has a bandwidth of 20 MHz, potentially covering multiple IoT communication channels, this allows concurrent authentication for multiple IoT devices via a single WiFi transmission. This requires the efficient allocation of spectral

and temporal resources to coordinate IoT transmissions without interfering with each other.

IV. AUTHFI IN A NUTSHELL

Fig. 2 presents an overview of AUTHFi, which leverages *commodity WiFi devices* to authenticate heterogeneous IoT devices (e.g., ZigBee and Bluetooth), in contrast to existing techniques based on SDRs. Without loss of generality, this paper focuses on the ZigBee scenario for illustration, while its design principles are generic and could be applied to other IoT devices (e.g., Bluetooth). The main design components in AUTHFi are illustrated as follows:

- **Waveform Reconstruction and Compensation:** AUTHFi leverages XFi's techniques [15] to analyze the received WiFi payload for partial IoT waveform recovery. The reconstructed ZigBee waveform suffers from inevitable signal losses due to LDPC and CP components. AUTHFi can compensate for CP loss part by interpolating using the intact portions within a single ZigBee symbol, ensuring more accurate reconstruction of the ZigBee preamble's waveform. This method is essential for accurate device authentication.
- **CFO Estimation:** To enable accurate CFO estimation under interference and noise, AUTHFi adjusts the existing CFO estimation techniques [6], [10]. First, it computes a weighted coarse CFO, considering the varying signal strengths of the reconstructed IoT waveform. Second, to refine the residual frequency offset, AUTHFi introduces a fine-grained CFO estimation using the Least Mean Squares (LMS) algorithm for enhanced calibration.
- **Device Authentication Model:** For accurate device authentication, we develop a new model that combines the CFO estimation results and sequential IQ values of the reconstructed waveform. It processes these inputs using an integration of an LSTM subnetwork and a fully connected subnetwork, culminating in a fusion network. Note that the LSTM is adopted for its ability to capture long-term dependencies in sequential data, making it suitable for analyzing time-series IQ values. This authentication network combines temporal signal patterns (IQ values) with the estimated CFO values to enable accurate device classification, even when the estimated CFO values fluctuate due to factors affecting the estimation algorithm. The network also incorporates rogue device detection for identifying unauthorized devices.
- **Multi-device Authentication:** The AUTHFi framework enhances the efficiency of authenticating multiple IoT devices via its coordination strategy. By efficiently leveraging the frequency and temporal resources, AUTHFi is able to verify multiple IoT devices with one single WiFi transmission.

V. THREAT MODEL

This section presents a comprehensive threat model for the AUTHFi system, outlining potential security risks and system vulnerabilities.

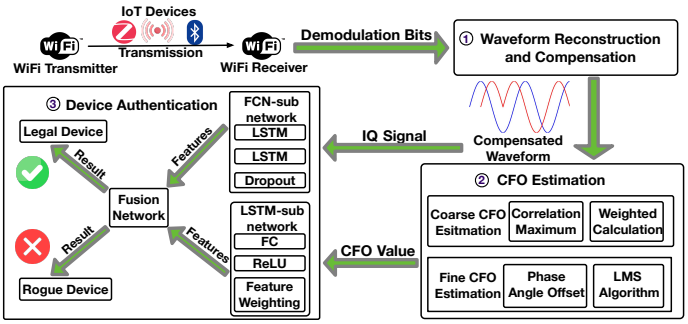


Figure 2: The framework of AUTHFi

A. Attacker Capabilities

1) **Signal Capture and Replay Attack:** The attacker can capture and replay communication signals from legitimate IoT devices. The attack method involves using SDR devices to record transmissions from ZigBee or Bluetooth devices, including physical layer characteristics, and then retransmitting these signals at a later time or different location [33]. The goal is to deceive the AUTHFi system into recognizing the replayed signal as a new transmission from a legitimate device.
2) **Man-in-the-Middle (MITM) Attack:** The attacker can intercept, modify, and forward messages between communicating parties. This attack may involve deploying malicious relay devices between IoT devices and WiFi receivers to manipulate transmitted signals in real-time [34]. The objective is to tamper with the authentication process, insert malicious instructions, or steal sensitive information.
3) **Device Cloning Attack:** The attacker attempts to replicate the identity information of legitimate IoT devices, such as MAC addresses, to impersonate them. This attack involves sniffing network traffic to obtain MAC addresses of legitimate devices and then configuring malicious devices to use the same MAC addresses [35]. The goal is to bypass identity-based access controls.
4) **Denial of Service (DoS) Attack:** The attacker can launch a large number of authentication requests to exhaust system resources. This attack may involve using multiple malicious devices to simultaneously send numerous authentication requests or employing signal interference techniques to disrupt normal communication processes [36]. The aim is to overload the authentication system and impair the authentication process for legitimate devices.
5) **Unknown Malicious Device Infiltration:** The attacker attempts to introduce unauthorized new devices into the network, bypassing existing authentication mechanisms. This attack may involve deploying devices with special hardware characteristics that attempt to generate physical layer features similar to legitimate devices. The goal is to evade AUTHFi's detection mechanisms and gain network access.
6) **Environmental Manipulation Attack:** The attacker tries to alter the communication environment to affect

signal propagation characteristics, thereby interfering with AUTHFi's authentication process. This attack may include introducing reflectors or absorbing materials to change multipath effects, or using directional antennas to alter signal strength distribution. The objective is to make it difficult for the AUTHFi system to accurately estimate CFO or analyze IQ samples, thus reducing authentication accuracy.

B. Attacker Limitations

1) **Difficulty in Replicating Physical Layer Features:** Attackers find it challenging to precisely replicate hardware characteristics such as CFO of devices. Since CFO stems from hardware imperfections and is influenced by multiple factors, making it difficult to accurately simulate at the software level.
2) **Constraints in Real-time Signal Processing:** Attackers face difficulties in modifying physical layer signal characteristics in real-time without introducing noticeable delays. Since there are some requirements for specialized hardware due to high-precision real-time signal processing and inherent latencies in such processes.
3) **Limitations in Environmental Control:** Attackers cannot fully control all environmental factors affecting signal propagation. Since indoor and outdoor environments are complex and variable, with multiple unpredictable sources of interference.

C. System Assumptions

1) **Uniqueness of Physical Layer Features:** Each legitimate IoT device possesses unique and hard-to-replicate physical layer characteristics.
2) **Environmental Dynamics:** The communication environment is dynamic, and these changes affect the physical layer characteristics of signals.
3) **Trustworthiness of WiFi Infrastructure:** The WiFi devices used to execute AUTHFi are trustworthy and not controlled by attackers.
4) **Security of Initial Training Data:** The initial dataset used for training the authentication model is secure and does not contain data from malicious devices.

VI. AUTHFi Design

This section elaborates on AUTHFi's design and focuses on ZigBee authentication for illustration purposes. First, AUTHFi reconstructs the partial waveform of IoT transmission and compensates for the signal losses in some CP parts. Then, it estimates the CFO values and incorporates the CFO estimation and IQ samples through an adaptive authentication model. Additionally, AUTHFi can be extended from ZigBee authentication to Bluetooth and multi-device authentication.

A. Waveform Reconstruction and Compensation

Leveraging XFi's techniques to analyze the received WiFi payload for partial IoT waveform recovery. The reconstructed ZigBee signal contains unavoidable gaps due to WiFi's LDPC and CP components, with approximately one-third of the signal (8 μs per 24 μs interval) missing based on WiFi specifications. These gaps, while necessary for WiFi operation, create discontinuities in the reconstructed IoT signal.

For signal losses caused by CP segments, which only span 0.8 μs and with our 4MHz sampling rate (0.25 μs per sample), we only need to recover 3-4 sampling points. This short time interval makes interpolation feasible and reliable. However, for LDPC-induced losses which span 3.2 μs (approximately 12-13 sampling points), such extended signal reconstruction would introduce significant errors. Therefore, we need to ensure that CP compensation is only performed when the CP loss region does not overlap with LDPC segments. To address these limitations, we have developed a new phase-aware signal reconstruction method that utilizes known standard signal characteristics. For a segment with reference points $t_{\text{ref},1}, ..., t_{\text{ref},N}$ and target points $t_{\text{target},1}, ..., t_{\text{target},M}$, where N represents the number of reference points and M the number of target points:

1) **Phase Change Rate Estimation from Reference Points:** The phase change rate $\dot{\theta}_{\text{ref}}$ is estimated using unwrapped phase values, taking advantage of the known standard signal's phase:

$$\dot{\theta}_{\text{ref}} = \frac{1}{t_{\text{ref},N} - t_{\text{ref},1}} \times [\text{unwrap}(\theta(t_{\text{ref},N}) - \arg[s(t_{\text{ref},N})]) - \text{unwrap}(\theta(t_{\text{ref},1}) - \arg[s(t_{\text{ref},1})])], \quad (1)$$

where $\theta(t)$ represents the phase of the received signal at time t, $\arg[s(t)]$ denotes the phase angle of the standard signal at time t, and the unwrap$(\cdot)$ function ensures phase continuity by unwrapping phase values.
2) **Reference Amplitude Calculation:** The reference amplitude A_{ref} is calculated as the mean magnitude of the received signal at reference points:

$$A_{\text{ref}} = \frac{1}{N} \sum_{k=1}^{N} |r(t_{\text{ref},k})|, \quad (2)$$

where $|r(t_{\text{ref},k})|$ represents the absolute amplitude of the received signal at the k-th reference point.
3) **Signal Compensation at Target Points:** The compensated signal $r_{\text{comp}}(t_{\text{target}})$ at target points is reconstructed using the reference amplitude, estimated phase rate, and known standard signal phase:

$$r_{\text{comp}}(t_{\text{target}}) = A_{\text{ref}} \cdot e^{j[\theta(t_{\text{ref},1}) - \arg[s(t_{\text{ref},1})] + \dot{\theta}_{\text{ref}}(t_{\text{target}} - t_{\text{ref},1}) + \arg[s(t_{\text{target}})]]}. \quad (3)$$

4) **Complete Compensated Signal:** The complete compensated signal is defined piecewise for both reference and target samples:

$$r_{\text{comp}}(t) = \begin{cases} r(t) & \text{for reference samples} \\ A_{\text{ref}} \cdot e^{j[\theta(t_{\text{ref},1}) - \arg[s(t_{\text{ref},1})] + \dot{\theta}_{\text{ref}}(t - t_{\text{ref},1}) + \arg[s(t)]]} & \text{for target samples.} \end{cases} \quad (4)$$

Fig. 3 illustrates the compensation process for ZigBee CP segments. In this process, known standard signal information is used as a reference to analyze phase characteristics and reconstruct missing signal segments caused by CP. However, this approach is ineffective for missing signal segments caused by LDPC signal, due to its extended duration.

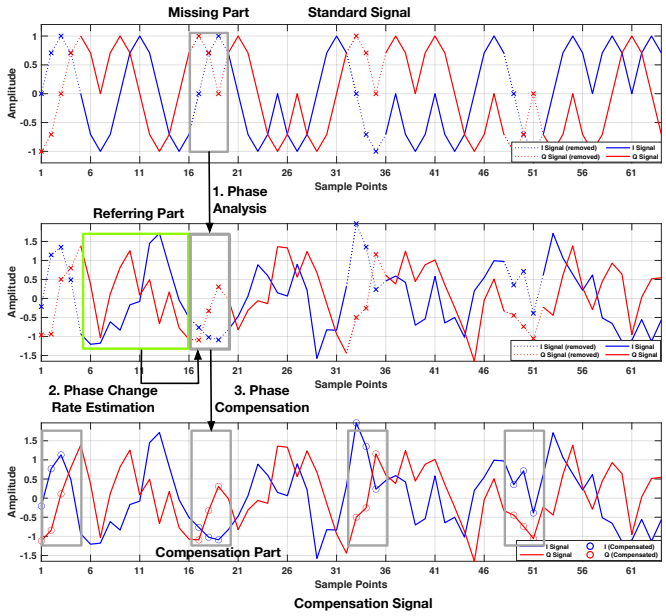


Figure 3: ZigBee CP compensation via reference phase analysis (short CPs only)

B. CFO Estimation

The hardware imperfection (e.g., unstable crystal oscillators) of IoT devices, leading to unique characteristics in the transmitted signal. Existing studies capture the IQ signal via SDR devices and further analyze it to extract the unique characteristics, such as CFO, for device authentication. This inspires us to compute CFO based on the reconstructed and compensated waveform. However, the calculated waveform suffers from noise and interference, having a low signal quality compared with the signal captured by SDR. As a result, AUTHFi tackles these issues and offers an accurate CFO estimation technique. (1) AUTHFi calculates the coarse CFO, leveraging existing correlation computation [25]. AUTHFi enhances reliability by introducing a weighted approach that considers the varying energy of the reconstructed waveform from the WiFi receiver. (2) Then, AUTHFi calculates the fine CFO to calibrate any residual frequency offset. In addition to the traditional method [10], AUTHFi employs the LMS algorithm to find a more accurate value.

C. Coarse CFO Estimation

Denote S^c as the compensated signal of the preamble. To ensure reliability and accuracy, AUTHFi introduces an optimization problem designed to precisely calculate the CFO. By maximizing the correlation between the preset signal sequence and the actual received signal, the optimal CFO value can be estimated. Specifically, the optimization problem is defined as follows:

$$\arg\max_{f_{\text{CFO}}^{\text{coarse}}} \sum_{n=1}^{N} \left| S^c[n] \cdot e^{-j2\pi f_{\text{CFO}}^{\text{coarse}} n T_s} \cdot S^{s*}[n] \right|, \tag{5}$$

where n is the index of the sampling point, T_s is the sampling interval (the reciprocal of the sampling rate), and N represents the number of samples for the ZigBee preamble with 8 symbols. The term $e^{j2\pi f_{\text{CFO}}^{\text{coarse}} n T_s}$ represents the phase rotation introduced by the CFO at each sampling point. The goal is to find the CFO value that maximizes the correlation between S^c and the standard preamble sequence S^s. Since it is difficult to find an analytical solution, AUTHFi adopts a numerical approach in practice. The numerical method tries different CFO values and evaluates the correlation change. The value that results in the highest correlation sum is selected as the coarse CFO. It will be further calibrated via fine CFO estimation for accuracy improvement.

Furthermore, we have observed that the reconstructed and compensated ZigBee signal has varying energy due to dynamic noises and the WiFi transmission's interference. Fig. 3 shows an example. To enhance the precision of the $f_{\text{CFO}}^{\text{coarse}}$ estimation, we introduce a weighted approach. Specifically, sampling points with larger amplitudes are given higher weights. Since these points tend to retain more useful ZigBee signal components compared with uniform weighting. The improved coarse CFO estimation formula is defined as follows:

$$\arg\max_{f_{\text{CFO}}^{\text{coarse}}} \sum_{n=1}^{N} \left| w[n] \cdot S^c[n] \cdot e^{-j2\pi f_{\text{CFO}}^{\text{coarse}} n T_s} \cdot S^{s*}[n] \right|, \tag{6}$$

where weight $w[n] = \frac{|S^c[n]|}{\sum_{m=1}^{N} |S^c[m]|}$ and m is the index of the sampling point. With the estimated $f_{\text{CFO}}^{\text{coarse}}$, AUTHFi adjusts the reconstructed signal as follows:

$$S^c[n] = S^c[n] \cdot e^{j2\pi f_{\text{CFO}}^{\text{coarse}} n T_s}. \tag{7}$$

D. Fine CFO Estimation

After coarse CFO estimation, a residual frequency offset still exists. To mitigate this, AUTHFi introduces fine CFO estimation. The calculation of the residual offset is facilitated by the periodic characteristics of the ZigBee preamble:

$$f_{\text{CFO}}^{\text{fine}} = -\frac{1}{2\pi L T_s} \cdot \angle \left(\sum_{n=1}^{N-L} S^c[n] \cdot S^{c*}[n+L] \right), \tag{8}$$

where L is the number of samples in one preamble symbol (i.e., $L = T_{\text{symbol}}/T_s$), and $\angle(\cdot)$ denotes the phase angle of a complex number. The summation runs from $n = 1$ to $N - L$ to ensure valid indexing within the signal length N.

While this method provides a preliminary estimation of the residual frequency offset, it may not achieve optimal accuracy in complex signal environments. In such cases, the LMS algorithm dynamically optimizes the CFO estimation by adjusting the estimation based on the deviation from the previous iteration, bringing the predicted CFO closer to the

actual value. This process helps mitigate residual frequency offset effects and improve signal quality. The corrected LMS algorithm is described as follows:

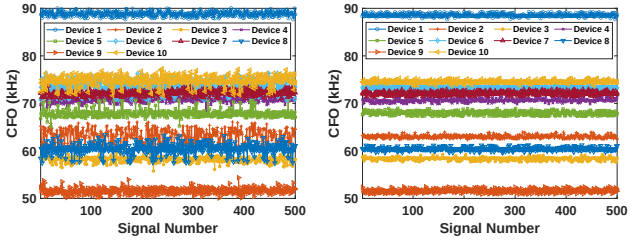
$$f_{\text{CFO}}^{\text{fine}}(i+1) = f_{\text{CFO}}^{\text{fine}}(i) + \frac{\mu}{2\pi L T_s} \sum_{n=1}^{N-L} \text{Im}\{S^c[n+L] \cdot S^{c*}[n] \cdot e^{-j2\pi f_{\text{CFO}}^{\text{fine}}(i) L T_s}\}, \tag{9}$$

where μ is the learning rate controlling the adjustment magnitude at each iteration, and $\text{Im}\{\cdot\}$ denotes the imaginary part of a complex number. The convergence condition typically requires that $0 < \mu < \frac{1}{\lambda_{\max}}$, where $\lambda_{\max}$ is the largest eigenvalue of the input signal's covariance matrix Σ_{S^c}.

$$\Sigma_{S^c} = \mathbb{E}\left[\left(S^c - \mathbb{E}\left[S^c\right]\right)\left(S^c - \mathbb{E}\left[S^c\right]\right)^T\right]. \tag{10}$$

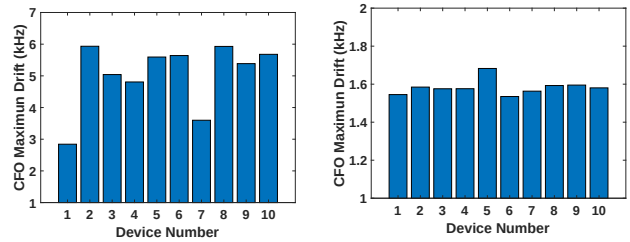
At last, AUTHFi computes the final CFO value as

$$f_{cfo}^{\text{final}} = f_{cfo}^{\text{coarse}} + f_{cfo}^{\text{fine}}. \tag{11}$$



(a) CFO fluctuation via existing methods (b) CFO fluctuation via AUTHFi

Figure 4: CFO fluctuations with time



(a) CFO drift via existing methods (b) CFO drift via AUTHFi

Figure 5: CFO drift comparison

To validate the CFO estimation reliability, our testbed includes 10 CC2652P USB dongles using ZigBee mode. Fig. 4 shows the fluctuation of the estimated CFO values with time. Via existing CFO calculation techniques [10], [25] (i.e., without AUTHFi's signal compensation and enhanced CFO calculation), the estimated CFO values suffer from significant fluctuation – different IoT devices might have overlapping CFO values. Notably, in Fig. 4(a), the maximum CFO drift can approach around 6 kHz in Fig. 5(a), which could adversely affect the authentication reliability. Via signal compensation and enhanced CFO calculation, AUTHFi significantly improves the CFO reliability in Fig. 4(b). Each device's CFO exhibits a clear boundary, with drift refined to within 1.7 kHz in Fig. 5(b). Moreover, our experiment found that fine estimation can reduce CFO drift by approximately 71.67% compared to using only simple coarse estimation. This reliable CFO estimation will serve as a solid foundation for reliable authentication.

E. Device Authentication Model

The CFO statistics can be derived for each device via the above method. However, the estimated CFO values may fluctuate due to multiple factors such as signal attenuation and multipath fading. To enhance detection accuracy, in Fig. 6, we propose a new network architecture for device authentication that integrates two types of inputs for robust identification: the sequential IQ from the preamble, and the CFO value [37]. Note that the training data consist of IQ and CFO values from M legitimate devices are reconstructed and calculated by AUTHFi, respectively. The IQ components are processed through LSTM layers to capture temporal signal patterns, followed by a dropout layer to combat overfitting. The CFO value is processed through a fully connected layer of 32 units, then through a ReLU activation, and finally through a 64-unit feature weighting layer that accentuates its role in the network's decision-making. Both streams merge in a fusion layer, combining their features, which are then processed by additional fully connected ReLU layers and dropout layers for generalization. The network culminates in an M+1-unit softmax output layer, corresponding to the device classes (i.e., including M legitimate and 1 rogue classes), yielding a probability distribution over potential device identities. The softmax function for the i-th class is given by:

$$P\left(y_i \mid \mathbf{x}\right) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}. \tag{12}$$

Here, $P\left(y_i \mid \mathbf{x}\right)$ is the probability of the i-th legitimate class given the input $\mathbf{x}$ (e.g., CFO or IQ value), z_i represents the confidence value for the i-th class, and K is the total number of classes. Moreover, to detect rogue devices, which are unauthorized devices that have not been seen in the training dataset, we define a decision metric $D(x)$ using the maximum class probability from the softmax function. A device is deemed rogue if this probability falls below P_{rogue}:

$$D(x) = \begin{cases} 1 & \text{if } \max(P(y_i|x)) < P_{rogue} \\ 0 & \text{otherwise.} \end{cases} \tag{13}$$

Note that $D(x) = 1$ flags a rogue device. The probability of the rogue device P_{rogue} is obtained by balancing the authentication of legitimate devices and the detection of rogue devices.

F. Bluetooth Extension

AUTHFi is a general framework that can be easily extended to Bluetooth authentication. Bluetooth employs Gaussian Frequency-Shift Keying (GFSK) for modulation, where data is transmitted through subtle frequency variations around its center frequency. Unlike the ZigBee protocol, Bluetooth does not incorporate a DSSS mapping process. Instead, the Bluetooth preamble consists of alternating 0 and 1 bits, providing an opportunity for signal compensation. Due to

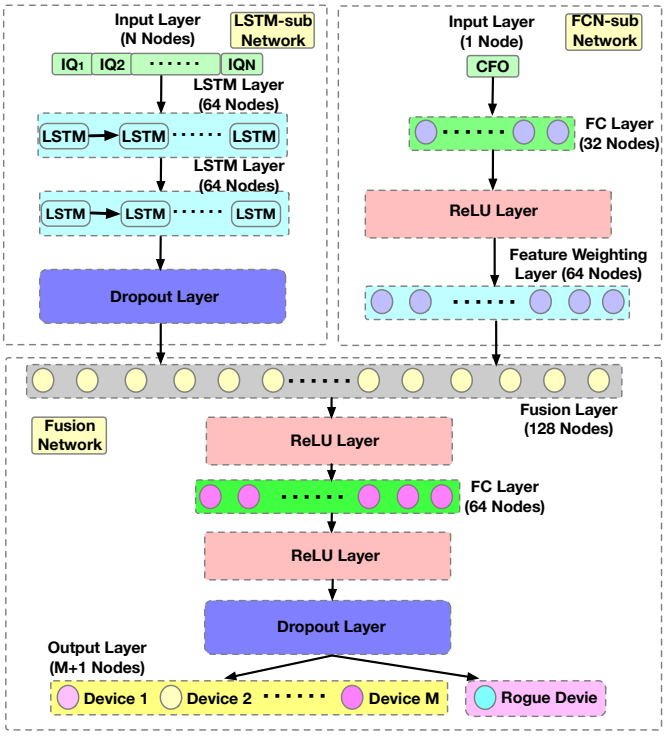


Figure 6: Neural network architecture for IoT device authentication with integrated IQ and CFO processing

the GFSK modulation characteristics, signals corresponding to 0 bits share consistent features, as do signals for 1 bits. Our compensation technique leverages this pattern consistency to reconstruct signals affected by CP loss in the preamble. By analyzing well-received segments, we establish reference patterns and calculate key parameters to maintain signal consistency. These reference patterns then guide the reconstruction of missing signal segments using linear interpolation method mentioned in Sec. VI-A while preserving the GFSK modulation characteristics.

G. Multi-Device Authentication

In this section, we extend our AUTHFi framework to efficient multi-device authentication by allocating frequency and time resources. AUTHFi leverages the bandwidth asymmetry between WiFi and low-end IoT devices (e.g., Bluetooth and ZigBee). As depicted in Fig. 7, a standard WiFi channel has a bandwidth of 20 MHz. This channel mainly consists of 48 subcarriers for data transmission, 4 pilot subcarriers for channel state estimation and 1 DC subcarrier. By setting the central frequency of the WiFi device to 2440 MHz, it is possible to use certain subcarriers (i.e., from 8 to 14 and 39 to 45) to establish two concurrent authentications for ZigBee devices at different channels. Since these two ZigBee channels do not overlap with each other, their transmissions will not lead to mutual interference. As a result, AUTHFi is able to reconstruct the signals of these two ZigBee transmissions by analyzing the received WiFi payload, followed by the techniques introduced in Section VI for authentication. Note that this channel allocation strategy can also be applied to Bluetooth protocol. In addition to frequency allocation, AUTHFi could also leverage time allocation to improve authentication efficiency. The idea is to allocate the communication slots of IoT devices within a single WiFi transmission. By doing so, AUTHFi enhances the authentication efficiency. Note that our multi-device authentication approach is mainly designed to authenticate multiple distinct devices simultaneously using different frequency channels. Moreover, we have collected and trained our model on CFO data from each device across its operational frequency range, accounting for frequency-dependent variations.

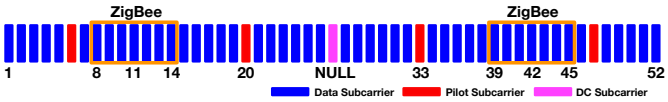


Figure 7: Channel allocation for multiple devices

VII. Experiment Evaluation

This section evaluates AUTHFi's prototype, utilizing commercial WiFi and IoT devices. Our evaluation covers CFO stability evaluation under various conditions, ZigBee/Bluetooth authentication accuracy, and rogue device detection.

A. Experiment Setup

In this work, we implement a prototype of AUTHFi using commercial off-the-shelf devices. Our setup, illustrated in Fig. 8, uses ALFA AWUS036ACH adapters with RTL8812AU chipsets (driver version 4.3.21) as WiFi transmitters and receivers. For IoT devices, we employ CC2652P USB dongles supporting both ZigBee and Bluetooth protocols, with applications developed using TI-RTOS SDK version 7.10.1.24. The core AUTHFi algorithms are implemented in MATLAB, utilizing WLAN Toolbox 3.5, Signal Processing Toolbox 9.4, and Deep Learning Toolbox 15.4. For verification, we use a USRP B210 with UHD driver 4.6.0.0 and GNU Radio 3.8. All experiments run on Ubuntu 16.04 LTS, with Wireshark 2.6.10 for packet analysis. We evaluate AUTHFi's performance across various scenarios, including CFO stability over time and distance, authentication accuracy of our fusion network, Bluetooth device authentication, and rogue device detection. This comprehensive setup enables a thorough assessment of AUTHFi's capabilities in real-world conditions.

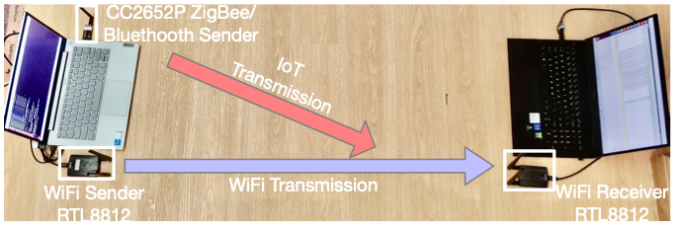
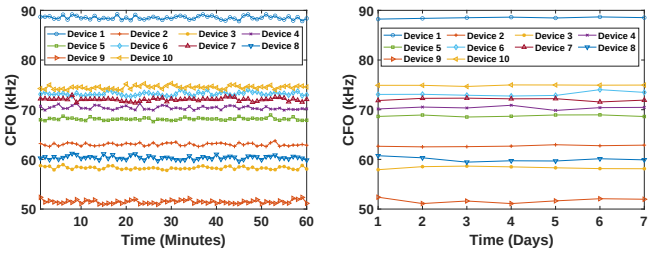


Figure 8: Experiment settings for AUTHFi

B. CFO Stability Evaluation

1) Temporal Variations in CFO: Fig. 9(a) presents an analysis of CFO for 10 ZigBee devices, where each device transmits a signal every minute over the span of one hour. The data reveals distinct CFO levels among the devices. Devices 3 and 9 show lower CFO values (i.e., 50-60 kHz) and Device 1 has a higher CFO (i.e., around 90 kHz), an indication of hardware diversity. Concurrently, Devices 6 and 7 display overlapping CFO values and fluctuation patterns, suggesting similarities in their frequency offsets. Subsequently, a long-duration CFO analysis is conducted over seven days, as shown in Fig. 9(b). This analysis involves transmitting 10,000 packets per day from each device and taking the mean. The results from Fig. 9(b) show stable and distinct frequency offsets for each device, confirming the presence of individual hardware characteristics. This consistent differentiation of CFO values over time highlights the potential of CFO as a reliable metric for device authentication within ZigBee networks. To further investigate the time-varying properties of CFOs, we measure CFO values from initial startup to 24 hours of continuous operation. As shown in Table II, the CFO values increase rapidly in the first 30 minutes before gradually stabilizing. The maximum CFO drift ranges from 2.8 to 4.1 kHz, with most devices stabilizing within 3 hours. While these warm-up period variations present a challenge, AUTHFi's fusion of CFO and IQ data in the authentication process helps mitigate the impact of initial CFO instability.



(a) One hour (b) One week

Figure 9: CFO variations for different ZigBee devices during different durations

Table II: CFO Measurements Over Time for Different Devices (kHz)

Dev	Init	30m	1h	3h	5h	12h	24h	Max
1	85.2	87.6	88.3	89.1	88.9	89.3	89.0	+4.1
2	60.1	62.0	62.3	63.1	62.8	63.2	63.0	+3.1
3	55.3	56.9	57.4	57.6	58.1	57.9	58.2	+2.9
4	69.2	70.5	71.3	71.6	72.1	71.8	72.2	+3.0
5	65.0	66.7	66.9	67.6	67.9	68.2	67.8	+3.2
6	72.1	73.8	74.0	74.6	75.0	74.8	75.1	+3.0
7	70.2	71.5	72.3	72.4	72.9	73.1	72.8	+2.9
8	58.1	59.7	60.0	60.6	60.9	61.0	61.2	+3.1
9	49.1	50.5	51.2	51.4	51.9	52.1	51.8	+3.0
10	73.3	74.7	75.3	75.5	76.0	75.8	76.1	+2.8

2) Distance-Induced Variations in CFO: The CFO values of IoT devices are assessed under varying distances and analyzed for the impact of Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) conditions. The WiFi receiver is placed at

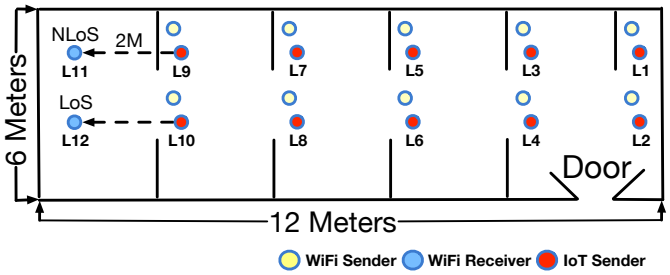


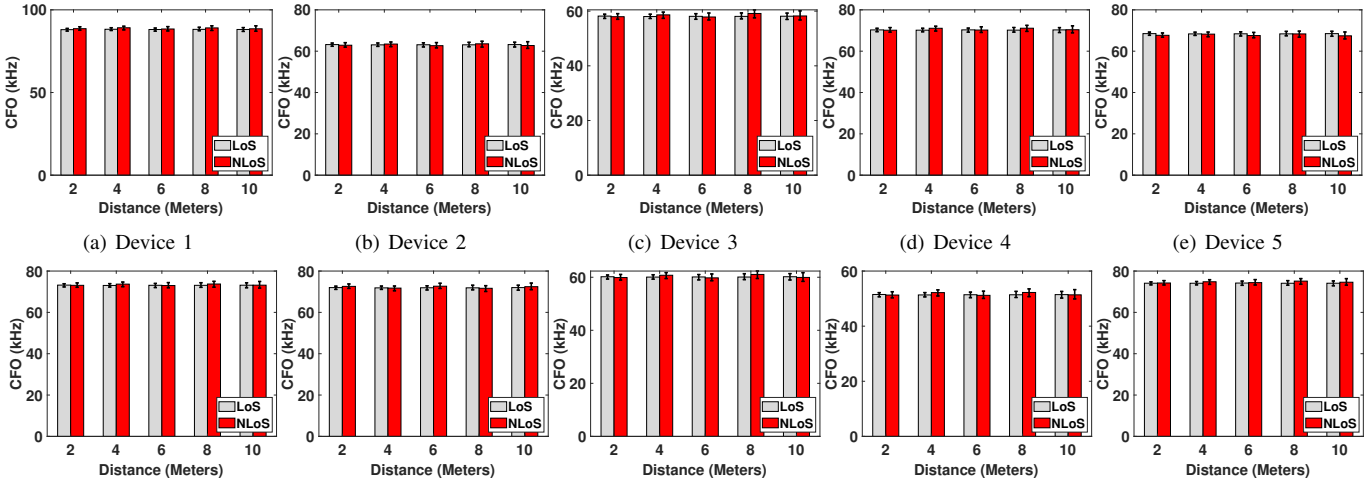
Figure 10: A university lab layout. Ten IoT transmitters are placed at locations L1 to L10. The WiFi receiver is stationed at either L11 or L12 to create LoS and NLoS environments. The WiFi transmitter is placed at every location from L1 to L10 for detailed examination

L11 or L12 to create LoS and NLoS environments. The WiFi transmitter is placed at every location from L1 to L10, as depicted in Fig. 10. Ten ZigBee transmitters are also placed at locations from L1 to L10. Locations L1, L3, ..., L11 are selected for NLoS measurements, while other locations for LoS measurements. The horizontal interval is set to 2 meters.

Fig. 11 illustrates that the estimated CFO values for each device maintain a level of consistency as the distance increases. These values are in line with our previous findings as shown in Fig. 9. However, the variability in the estimated CFO grows with increased distance, particularly in NLoS scenarios compared to LoS. The greater fluctuations in NLoS are attributed to obstructions that disrupt the signal, which can affect the accuracy of the CFO estimation algorithm. It is important to note that these fluctuations are in the estimated CFO values and not the true CFO, which is determined by the difference between two frequency oscillators. Note that while RSSI could provide additional insights, distance is chosen as the x-axis to facilitate direct comparison between LoS and NLoS scenarios, where RSSI values may differ significantly at the same distance due to environmental factors.

C. Authentication Reliability

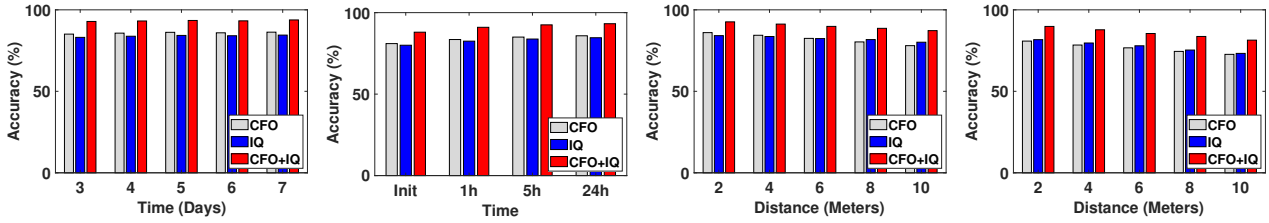
1) Temporal Variations in Authentication Performance: we evaluate AUTHFi's fusion network which integrates Fully Connected Network (FCN) and the LSTM network. This network integrates both CFO values and preamble IQ signals, as depicted in Fig. 6. In addition, we also evaluate the authentication performance of the following two individual networks: FCN network (based on CFO estimation) and LSTM network (based on IQ values). The training dataset includes ZigBee packets from the first two days, shown in Fig. 9. The testing is conducted using ZigBee packets collected in the five consecutive days and the packets under various distances and coordination schemes. During the remaining five days, as illustrated in Fig. 12(a), the CFO network consistently outperforms the IQ network, with accuracy between 85.1% to 86.3%, compared with 83.1% to 84.5%, respectively. The CFO method demonstrates greater consistency over time, whereas the IQ signal is more variable and less stable. This indicates that the IQ signal exhibits more fluctuation under time variations compared to the CFO method. Notably,



(a) Device 1 (b) Device 2 (c) Device 3 (d) Device 4 (e) Device 5

(f) Device 6 (g) Device 7 (h) Device 8 (i) Device 9 (j) Device 10

Figure 11: CFO values for ZigBee devices under different distance for N/LoS scenario



(a) Days 3-7 (b) Warm up period (c) LoS (d) NLoS

Figure 12: Authentication accuracy for ZigBee devices under different settings

the CFO+IQ network achieves the best performance on the subsequent five days of data, maintaining accuracy between 92.87% to 93.80%. Although both the CFO network and IQ network show slightly lower accuracy compared to the fusion approach, their performance remains stable across the testing period. Therefore, the combined CFO+IQ network can effectively address the shortcomings of a single network in the authentication process, improving accuracy by approximately 7-8%.

Moreover, Fig. 12(b) illustrates the authentication accuracy during the warm-up period of devices. The CFO+IQ fusion network consistently outperforms individual networks, achieving 93.2% accuracy after 24 hours. This demonstrates the fusion network's robustness to initial CFO instabilities. Notably, the CFO network's performance improves over time, from 81% to 85.8%, as the device's frequency stabilizes. The IQ network shows a similar trend, improving from 80% to 84.6%, while the fusion network leverages the complementary strengths of both approaches to achieve accuracy improvements from 88% to 93.2%.

2) Distance-Induced Variations in Authentication Performance: Under the LoS conditions, the CFO network accuracy declines with increasing distance, falling to about 78% at 10 meters as shown in Fig. 12(c). Similarly, the IQ network drops to 80.1%, while the CFO+IQ network maintains higher accuracy at 87.2%. Under the NLoS conditions, single CFO and IQ networks show more obvious limitations due to environmental obstructions, as Fig. 12(d) indicates a 13.3% and 10.9% drop in their performance, respectively. However, the CFO+IQ network remains more stable with accuracy around 81.4% at 10 meters. Overall, the CFO+IQ network outperforms individual networks by an average of 8.2%, owing to its integrated processing of two physical-layer features (i.e., CFO and IQ), capturing a broader range of data characteristics for more reliable device authentication.

Table III: Comparison of Authentication Methods

Category	Method	Accuracy (%)	FPR (%)
Conventional	Threshold-based CFO	78.5	9.8
	SVM	84.2	7.3
Deep Learning	CNN-only Approach	89.1	5.6
	LSTM-only Approach	91.3	4.5
	AUTHFi (Ours)	**93.5**	**3.5**

3) Different methods in Authentication Performance: To demonstrate the effectiveness of our deep learning-based approach, we compare AUTHFi with both conventional methods and other deep learning architectures using data from the third day of our experiment. As shown in Table III, conventional methods such as threshold-based CFO (using a simple threshold on CFO values) and Support Vector Machine (SVM) (applying SVM on CFO features) achieve 78.5% and 84.2% accuracy with 9.8% and 7.3% False Positive Rate (FPR), respectively. In contrast, deep learning methods show better performance. A CNN-only approach using IQ samples achieves 89.1% accuracy with a 5.6% FPR, while an LSTM-

only network on IQ samples reaches 91.3% accuracy with a 4.5% FPR. Our AUTHFi method, which combines CFO and IQ data using a fusion of CNN and LSTM architectures, outperforms all others with 93.5% accuracy and only a 3.5% FPR.

Table IV: Authentication Accuracy in Mobile Scenarios

Mobile Scenario	Accuracy
Static (Reference)	88-90%
Slow Movement (Walking at 2-6 km/h)	82-88%
Medium Movement (Fast Walking/Slow Jogging at 6-10 km/h)	76-85%
Fast Movement (Running at 10-12 km/h)	65-70%
Extreme Cases (Quick Turning/Obstacle Crossing)	58-65%

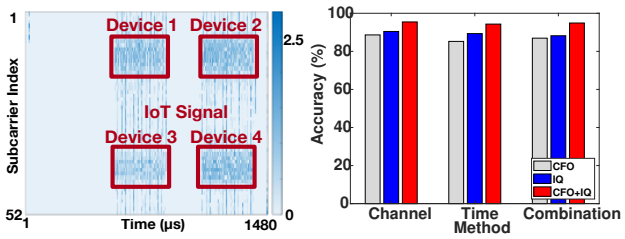
4) Mobility in Authentication Performance: Table IV presents the results of our authentication system's performance under various mobility scenarios, simulated using a motorized track in both indoor and outdoor environments. In static conditions, the system demonstrates exceptional accuracy, ranging from 88% to 90%, with variations attributed to distance and LoS factors. As mobility increases, introducing Doppler effects, the system's performance exhibits a gradual decline while maintaining robust functionality. At walking speeds (2-6 km/h), the system sustains high accuracy levels of 82-88%. As speed increases to fast walking or slow jogging (6-10 km/h), where Doppler effects become more pronounced, accuracy remains impressive, ranging from 76% to 85%. Notably, even at running speeds (10-12 km/h), which induce significant Doppler shifts, AUTHFi maintains a commendable accuracy of 65-70%. In extreme scenarios, such as sharp turns that cause sudden channel changes, the system's accuracy experiences a more substantial drop, ranging from 58% to 65%.

D. Concurrent Authentication Accuracy

This section validates the multi-device authentication in AUTHFi. Each transmitted WiFi packet includes a notification via CTC, which informs specific IoT devices about their assigned transmission channels and time. Upon receiving this CTC, IoT devices start to transmit following the CTC packet's allocation. AUTHFi utilizes the techniques described in Section VI to analyze the received WiFi payload for IoT device authentication. Fig. 13(a) displays the Time-Frequency usage within the WiFi transmission. We can observe four distinct signals that originate from Devices 1 to 4. We evaluate the authentication of using time allocation, channel allocation, and the combined approach. The authentication results are presented in Fig. 13(b), further underscores AUTHFi's reliable authentication even in a scenario of authenticating multiple devices. Similar to Fig. 12, the CFO+IQ fusion neural network accomplishes the best performance, achieving approximately 8% and 9% accuracy increase than the individual CFO and IQ networks in each scenario.

E. CFO Estimation Verification

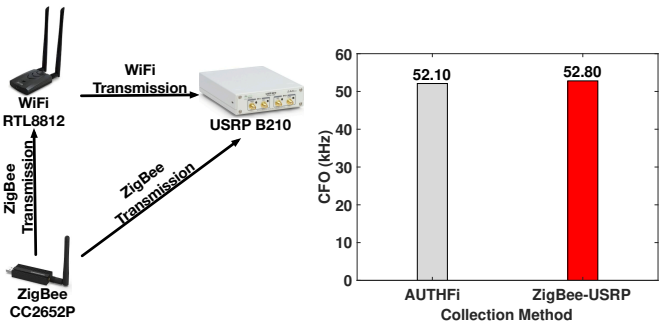
Here, we verify CFO estimation accuracy of AUTHFi. To achieve this, we utilize a USRP B210 SDR for capturing signals transmitted by the IoT device and the WiFi source as shown in Fig. 14(a). First, we leverage AUTHFi to calculate



(a) Subcarrier utilization (b) Authentication accuracy

Figure 13: Concurrent authentication of multiple IoT devices via coordination

the ZigBee device's CFO, following the design in Section VI. Then, we leverage the USRP to calculate the CFO of the ZigBee and WiFi devices, respectively. The ZigBee device's CFO is calculated following our design in Section VI, while the WiFi device's CFO is calculated using the short and long training sequence module provided in the gr-ieee802-11 PHY [38]. The calculated CFO for the ZigBee device is approximately 83.4 kHz when measured using USRP, whereas the WiFi device exhibits an average CFO of around 30.6 kHz under the same USRP measurement condition. Then, we calibrate the ZigBee device's CFO estimation based on USRP by deducting the WiFi device's CFO measured on USRP. The calibrated value is 83.4 - 30.6 = 52.8 kHz. In Fig. 14, the CFO values measured by AUTHFi and USRP closely approximate each other. Hence, the CFO estimation of AUTHFi has been verified.

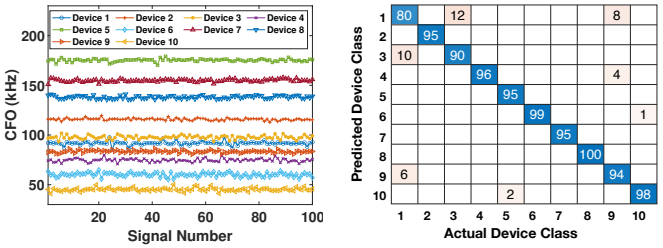


(a) Calibration scenarios (b) CFO value comparison

Figure 14: Verification of AUTHFi's CFO estimation

F. Bluetooth Extension

In this experiment, we use Bluetooth chips on 10 CC2652P devices to test the performance of our AUTHFi system. Note CC2652P devices contain both ZigBee and Bluetooth chips. The results, as illustrated in Fig. 15(a), show reliable CFO patterns ranging from 40 kHz to 180 kHz across the devices. Notably, there is some overlap between Device 3 and Device 1 around 95 kHz. However, this issue has been successfully addressed by our improved CFO+IQ authentication model. The effectiveness of this approach is further evidenced in Fig. 15(b), where the confusion matrix displays an impressive average classification accuracy of 94.2%.

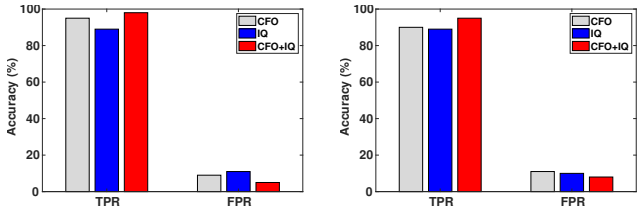


(a) CFO for different Bluetooth devices (b) Confusion matrix for Bluetooth device authentication

Figure 15: Bluetooth CFO and authentication performance

G. Rogue Device Detection

In addition to authenticating legitimate devices, we evaluate AUTHFi's performance of detecting rogue devices. Note that rogue devices refer to an unauthorized device and has not been seen in the training dataset. To evaluate this aspect of performance, we utilize two key metrics–True Positive Rate (TPR) and FPR. The TPR quantifies the percentage of legitimate device signals that AUTHFi accurately identifies, whereas the FPR measures the percentage of adversarial signals that AUTHFi mistakenly recognizes as legitimate. For AUTHFi performance evaluation, we also evaluate the above-mentioned three models (i.e., FCN, LSTM and fusion models). Furthermore, our training dataset comprises ZigBee and Bluetooth signals gathered from 10 CC2652P devices, while our testing set includes data from two additional CC2652P devices that are not part of the training phase. Regarding ZigBee rogue device detection, the CFO+IQ model performs well, achieving a 98% TPR and a 5% FPR in Fig. 16(a), indicating its good ability to accurately identify legitimate devices while minimizing false identifications. In the case of BLE, the CFO+IQ model similarly stands out with a high TPR of 95% and a relatively low FPR of 8% as shown in Fig. 16(b). The experiment results confirm our AUTHFi effectiveness in both correctly detecting legitimate devices and reducing misclassifications compared to the individual CFO and IQ models.



(a) Rogue ZigBee devices (b) Rogue Bluetooth devices

Figure 16: Rogue device detection performance for AUTHFi

H. Complexity Analysis

We conduct a comprehensive theoretical analysis of AUTHFi's efficiency, focusing on authentication complexity and processing time. Our analysis is based on the ZigBee 2 MHz bandwidth standard configuration, using 512 sampling points (256 chips with 2x oversampling, 4 MHz sampling rate). We base our analysis on common embedded WiFi devices, such as those using a 710 MHz Qualcomm IPQ4019 processor[1].

- **Signal Reconstruction and Compensation:** The complexity of this phase is O(N), where N = 512. Each sample requires 10-15 basic operations, totaling 7,680 operations. The theoretical execution time is approximately 0.011 ms.
- **CFO Estimation:** This process consists of two parts:
  1) Coarse estimation using FFT: The complexity is O(NlogN) $\approx$ O(4608). Since the FFT operation requires 23,040 complex operations, equivalent to 92,160 real operations [39]. Theoretical execution time is about 0.130 ms.
  2) Fine estimation using LMS algorithm: The complexity is O(N). Assume 10 iterations, each requiring 30 basic operations, we total 153,600 operations [40]. Theoretical execution time is about 0.216 ms.
- **Neural Network Inference:** With approximately M=100,000 parameters, the complexity is O(M). Estimating 1,000,000 total operations (including LSTM layer complexity), the theoretical execution time is about 1.408 ms.
- **Overall Analysis:** The total theoretical time complexity is O(NlogN + M), where N = 512 and M $\approx$ 100,000. Hence, we can estimate the total processing time by summing the individual components: Total Time = Reconstruction + CFO Estimation + NN Inference = 0.011 + (0.130 + 0.216) + 1.408 = 1.765 ms. To validate our analysis, we conduct offline tests on a laptop with an Apple M1 Pro processor. Results show a total processing time of around 1.5 ms, aligning well with our theoretical estimates. The slight improvement in actual performance can be attributed to the M1 Pro's powerful neural network accelerator and efficient memory architecture [41].

VIII. Security Analysis

This section analyzes the security of the AUTHFi system based on the previously defined threat model.

A. Defense Against Signal Replay Attacks

AUTHFi utilizes CFO and IQ samples as physical layer features of devices. These features are inherently time-varying and influenced by environmental factors. AUTHFi's authentication model considers the temporal patterns of IQ samples. The replay attack would superimpose a new temporal pattern onto the original one: the pattern of the original signal and the pattern introduced by the replay process. This composite pattern would significantly differ from the expected pattern of a legitimate device [42]. For instance, if an attacker replays the same signal at different times, environmental changes would cause inconsistent temporal patterns in the IQ samples, which can be detected by AUTHFi.

[1]https://github.com/Deoptim/atheros/blob/master/IPQ4019-datasheet.pdf

B. Mitigation of MITM Attacks

AUTHFi directly utilizes physical layer features for authentication, not relying on higher-layer protocol data that could be intercepted and tampered. Higher-layer protocol data refers to information exchanged at network and application layers, such as packet headers, encryption keys, or authentication tokens. These data are typically easier for attackers to intercept and manipulate compared to physical layer characteristics. By basing authentication on physical layer features, AUTHFi makes it significantly more challenging for attackers to successfully execute a MITM attack. Even if an attacker can intercept the communication, it is difficult to accurately replicate or modify physical layer features in real-time [43]. The fusion network creates a complex signature that is challenging for an attacker to manipulate coherently. Furthermore, AUTHFi's authentication process does not involve the exchange of sensitive information, reducing the risk of MITM attacks.

C. Resistance to Device Cloning Attacks

AUTHFi's authentication relies not only on device identification information but also on physical layer features that are difficult to clone. Even if an attacker successfully clones a device's MAC address or other identifying information, they cannot replicate its unique CFO characteristics, as CFO is determined by the device's hardware properties [42]. As shown in Fig. 4 and 5 of the paper, CFO values differ significantly between devices, enabling CFO-based authentication to effectively identify cloned devices.

D. Mitigation of DoS Attacks

AUTHFi can authenticate multiple devices simultaneously and utilizes existing WiFi infrastructure for efficient authentication. The capability for concurrent authentication helps maintain authentication functionality under high traffic conditions. Leveraging existing WiFi infrastructure allows for rapid processing of authentication requests, reducing the risk of resource exhaustion. As demonstrated in Fig. 13 of the paper, AUTHFi can authenticate multiple devices simultaneously, enhancing the system's overall throughput and resistance to DoS attacks.

E. Detection of Unknown Malicious Devices

AUTHFi's authentication model includes components specifically designed to detect malicious devices. By setting a probability threshold (P_{rogue}), the system can identify devices not seen in the training dataset. As shown in Fig. 16 of the paper, AUTHFi achieves high true positive rates (98% for ZigBee, 95% for Bluetooth) and low false positive rates (5% for ZigBee, 8% for Bluetooth) in detecting malicious devices.

F. Environmental Adaptability

AUTHFi's fusion network demonstrates high accuracy across various environmental conditions. The fusion network can adapt to different signal propagation environments, including NLoS conditions. As illustrated in Fig. 12(d) of the paper, even under NLoS conditions, the fusion network maintains an accuracy of around 80-90%, indicating its strong adaptability to environmental changes. This security analysis comprehensively evaluates the performance of the AUTHFi system in the face of various attacks, demonstrating its effectiveness in addressing different security threats.

IX. Related Work

In this section, we discuss recent developments in using physical-layer features like IQ offset and CFO for IoT device authentication. Then, we explore some work for CTC, covering aspects such as packet and physical level CTC, the security issues in CTC, and potential countermeasures.

A. Device Authentication

In the dynamic field of IoT security, ensuring secure communication through device authentication is crucial, particularly in resource-limited settings. To achieve this goal, cryptographic protocols are widely used for IoT security [44]. Moreover, [45] also proposes a user authentication scheme for cloud-assisted IoT systems. However, these cryptographic methods often face limitations due to their considerable computational overhead and resource usage, which are unsuitable for low-end IoT devices that have tiny batteries.

To address the limitations in encryption, researchers have explored the unique characteristics of physical-layer signals for recognizing authorized nodes. The physical layer features that are inherently difficult to alter or counterfeit. For example, I/Q offset is explored in [46], where advanced deep learning techniques are applied to enhance Radio Frequency Fingerprinting (RFFP) for authenticating LoRa devices at the physical layer. Similarly, [27] delves into different signal representations for RFFP, and [47] investigates the 'Day-After-Tomorrow' effect in RFFP and its implications for signal authentication.

Meanwhile, CFO is another physical layer feature caused by the imperfection of the low-cost crystal oscillators embedded in IoT nodes. For instance, [6] and [10] demonstrate how CFO can significantly improve authentication accuracy for ZigBee and LoRa devices, respectively. Additionally, [8] examines the complexities of fingerprinting mobile devices using Bluetooth, highlighting how to enhance CFO-based physical-layer identification. However, the above-mentioned studies often employ SDRs for collecting IQ samples, which is the foundation of signal analysis. In contrast, our AUTHFi method uses commodity WiFi devices to capture IQ signals of IoT devices, offering a cost-effective solution for IoT device authentication.

B. CTC Techniques

CTC is a new approach that enables direct data exchange between devices using different communication technologies without hardware modification. It bridges the gap among heterogeneous wireless systems, enhancing interoperability and enabling more integrated network ecosystems [24].

The initial works in packet level CTC focus on packet size optimization for transmission such as [48] and [49]. Then subsequent packet-level CTC works utilize packet energy and

existence for interference mitigation and throughput improvement. For example, WiZig [50] optimizes data throughput in noisy channels using a theoretical model and rate adaptation algorithm, while StripComm [51] introduces an interference-resilient coding scheme to enhance CTC efficiency and reliability in crowded wireless environments. Despite advancements, packet-level CTC faces limitations like low throughput [23]. Then, CTC has evolved from packet-level approaches to physical-level methods that offer much better throughput.

For example, [12] introduces direct WiFi to ZigBee communication through WiFi modulation alteration. This leads to further physical-level CTC developments [13] and [14], enhancing WiFi to Bluetooth and LoRa communications. Additionally, there has been growth in low-end transmitter CTC schemes, with notable developments [22] and [15] in Bluetooth to ZigBee and LoRa/ZigBee to WiFi communications.

While CTC offers convenience, it also introduces new security challenges. [52], [53] have claimed that CTC can enable more concealed wireless jamming attacks. Hence, [52] proposes a defense mechanism that detects CTC. Moreover, [54], [55] focus on embedding and verifying authorization codes in packet preambles and analyzing physical layer traits for verifying the CTC transmitter. However, existing work commonly focuses on detecting the existence of CTC or verifying the legitimate CTC sender. Moreover, this often involves the use of costly USRPs for data collection.

In contrast, our AUTHFi aims for general authentication of IoT devices by leveraging ubiquitous WiFi devices. AUTHFi inherently differs from existing solutions and requires addressing several unique challenges.

X. Conclusion

In this paper, we present AUTHFi, a novel cross-technology device authentication framework that enables commodity WiFi devices to authenticate IoT devices such as ZigBee and Bluetooth. In contrast to prior solutions that rely on expensive SDRs, AUTHFi leverages recent advances in CTC to reconstruct partial waveforms of IoT transmissions using standard WiFi devices. To achieve accurate authentication, AUTHFi introduces several key innovations. First, it compensates for partial signal losses in the reconstructed waveform by using linear interpolation on the intact portions of the signal. Second, it enhances CFO estimation under low signal quality through weighted coarse CFO calculation and fine-grained calibration. Third, AUTHFi develops a fusion neural network that combines the CFO and reconstructed waveform's IQ values for robust authentication in the face of environmental variations. Finally, it enables efficient concurrent authentication of multiple IoT devices by optimizing frequency and temporal resource allocation. In future work, we plan to explore the integration of advanced signal processing and machine learning techniques to bolster the robustness of AUTHFi in environments with substantial interference.

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